Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2507505

Unveiling the Timeworn Scars and Knowledge of Stairs

Summary

Stair is significant in our daily life as a fundamental architectural element. The wear, which is influenced by different material properties, usage frequency and usage preferences, provides valuable insights into a stairwell's history and usage patterns. However, interpreting the wear patterns can be complicated by factors like repair and renovations and material changes. Developing multi-materials universal models can help archaeologists estimate the age of any stairs, infer usage patterns and determine the number of simultaneous users, offering deeper insights into the daily activities of past inhabitants.

Several models are established: **1) SEMA Model, 2) SEFAM Model, 3) SUP-PFM Model.**

For Task 1, we develop a data measurement and tool preparation plan based on requirements while effectively controlling the budget. Specific methods for verification were tailored according to the characteristics of rigid materials and wood. We can also acquire the frequency of repair. Subsequently, the acquired point cloud data was used for **3D surface reconstruction**, and the wear volume was calculated by **Delaunay triangulation** to quantify the extent of wear.

For Task 2, we solve problems of Usage Frequency, Age Estimation and Temporal Distribution. First, we develop **Strain Process and Stress Distribution Analysis** to: (1) simulate stair structures to discretize individual element using Finite Element Analysis (FEA), and (2) investigate stress, deformation, and dynamics. For **Usage Frequency**, we integrate Hertzian Theory, **Archard Equation** and Naismith's Rule, incorporating environmental noise, to estimate usage frequency **(SEMA Model)**, accounting for repairs by accumulating wear cycles from each repair instance; For **Age Estimation**, we combine **Poisson distribution** and Weathering Rind Logistic model through weighted averaging to estimate age **(SEFAM Model)**. The accuracy of SEFAM exceeds 96% across all materials; For **Temporal Distribution**, we estimate daily usage and apply Miner's Rule to assess usage intensity.

For Task 3, we solove the problems of the usage patterns of the stair. We analyze wear distribution using 3D image, integrating mechanical analysis with anthropometric measurements. For **Direction Preferences**, we infer primary usage (go upstairs/downstairs) and pedestrian side preferences (left/right). For **Side-by-side Preferences**, we estimate the number of total walkers stand on one step simultaneously based on physiology. **(SUP-PFM Model)**

For Task 4, **Sensitivity Analysis** using **Monte Carlo simulation** with 5–7% parameter oscillation shows: in SEMA, load frequency most impacts wear (24.54% increase); in SEFAM, temperature and humidity significantly affect wear ($\Delta W = 2.24 \times 10^{-4}\,\mathrm{m}^3$ and $\Delta W = 3.32 \times 10^{-4}\,\mathrm{m}^3$), while weathering and restoration have minimal effects.

Keywords: Finite Element Analysis, Pedestrian Flow, Archard Equation, Poisson Distribution, Fatigue Wear

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1 Introduction

1.1 Background and Restatement of the Problem

Stairs, as a common architectural element in everyday life, undergo wear over time due to constant human traffic. There are many ancient and incompletely documented staircases require our attention. The wear can provide valuable insights into the historical usage patterns, construction techniques, and even the cultural significance of a structure.

Our study aims to develop a mathematical model to analyze stair wear patterns, enabling archaeologists to draw conclusions about the history and usage patterns of staircases in ancient structures. We need to solve the following problems:

Task 1: Develop data collection and processing plans, including data types, tools, and budget. Confirm staircase material types and repair frequency, then preprocess the data.

Task 2: Construct a multi-material universal model for **Usage Frequency**(The amount of people per day) and **Age Estimation**. We need to consider the times of repair or renovations during this part. Analyze known data to assess **Temporal Distribution** (high traffic short-term vs. low traffic long-term).

Task 3: Discover the **Usage Pattern** of the stair. First, we need to identify the **favored direction** of movement for people (**Both horizontal and vertical**). Second, we need to identify whether people in this area usually walk alone or in groups and confirm the number of people (**Side-by-side Preferences**).

Task 4: Test our model and perform **sensitivity analysis**.

1.2 Literature Review

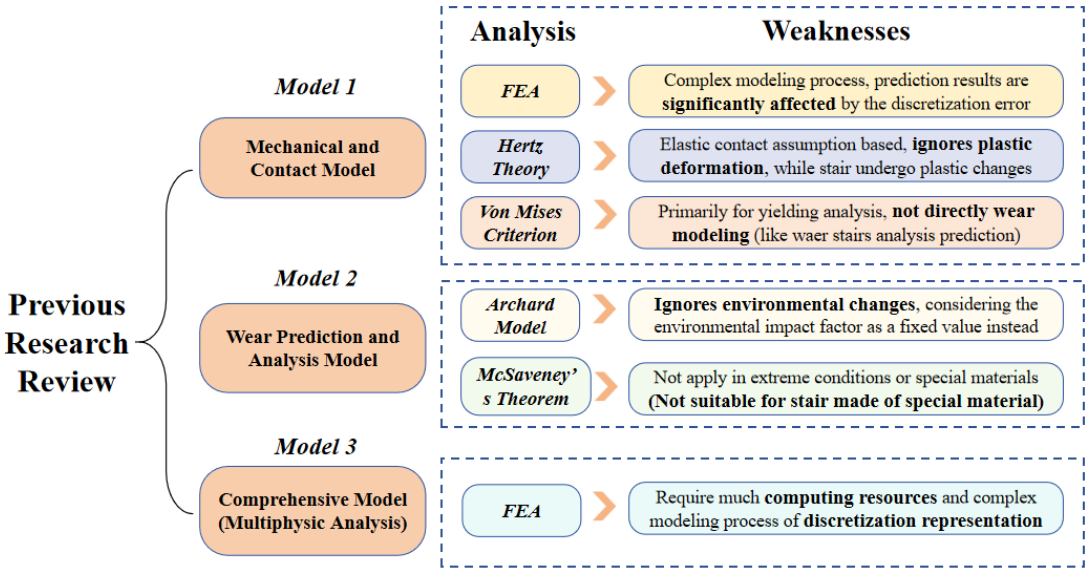


Figure 1: Literature Review

1.3 Our Work

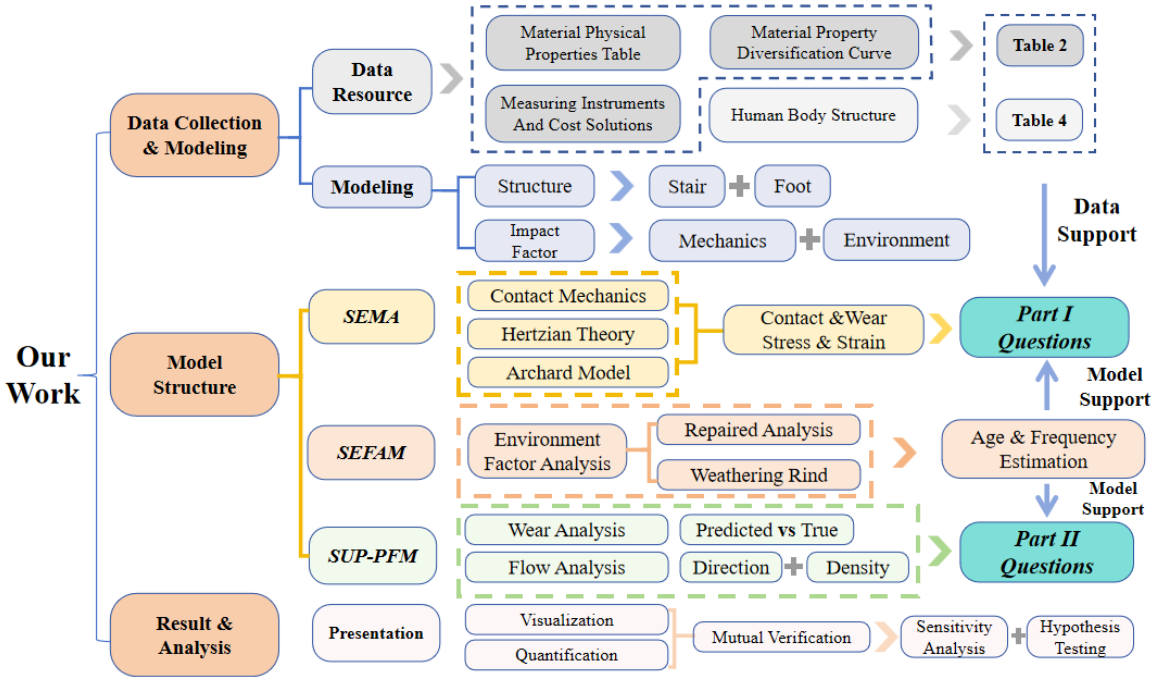


Figure 2: Structure of Our Work

2 Assumptions and Nomenclature

2.1 Assumptions

Assuption 1: We assume that the hardness of the staircase building material does not change, but we combine environmental noise to fit the model, avoiding certain errors.

Assuption 2: The wear of stairs is mainly caused by the normal natural environment (physical, chemical, and biological factors) and the mechanical wear of pedestrians, without considering extreme natural disasters or man-made damage.

Assuption 3: In the crowd flow simulation, high-density gatherings under special activities are not considered. It is assumed that the flow of people is large under good natural environment and suitable stair conditions (such as moderate slope and less damage).

Assuption 4: It is assumed that the speed of pedestrians when going up the stairs decays logarithmically, and a control coefficient is introduced, which leads to an increase in speed decay in the later stage.

Assuption 5: Suppose that the mode of the pedestrian walking on the stairs follows the most scientific mode.

2.2 Nomenclature

Table 1: Major Notations

Symbol	Description	Unit
h	The height of the one step	m
d	The depth of the one step	m
l	The width of the one step	m
W	Wear	m^3
K	Stiffness	N/m
H	Hardness	kgf/mm^2
E	Elasticity modulus	Pa
L	Load	N
Q	Wear radius	$\%$
A	Age of stairs	year
k, k(t)	Wear coefficient	/
m	The average weight of human	kg
N	Usage frequency	times
b(t)	Environmental impact	/
q	Momentum	$kg \cdot m/s$
g	Gravity Acceleration	m/s^2

3 Data Collection and Processing

3.1 Data Collection

Here is a table outlining the necessary data measurements to be conducted, along with an estimation of the associated budget. Next, we selected the key material properties (Table 2) to facilitate the analysis of the relationship between material stress and strain, and to optimize the **cost-efficiency** while ensuring **compliance with the measurement standards** to the greatest extent possible.

Table 2: Measurement Attributes and Costs

Attribute and Property (Usage)	**Measurement**	**Cost ($)**
Step Width (Shape detect)	Caliper (precise)	5–15
Step Depth (Shape detect)	Laser Distance Meter	15–40
Step Height (Shape detect)	Spirit level (basic)	10–20
Material (Material Verification for Rigid Materials)	X-Ray Fluorescence (XRF)	10–20
Load Value (Analyze the wear of stairs)	Portable load cell	15–25
Stress & Strain (Analyze wear and force)	Strain gauge	10–30
Stiffness (Analyze deformation and force)	Direct loading Method	100–150
Hardness (Analyze relevance of deformation & force)	Ultrasonic Hardness Tester	2,000
Elasticity Modulus (Analyze deformation and force)	Tensile testing module	50–200
Poisson's Ratio (Analyze material's impact on stair)	Strain gauge method	150–300
Thickness-Reflectance Intensity (confirm number of repairs)	Ultrasonic Thickness Gauge	100–300
Stair material distribution (Repair and Renovations Situation)	Handheld LiDAR	300-500

3.2 Material Verification

3.2.1 Material Verification for Rigid Materials

By comparing the elemental composition obtained from XRF analysis with known material databases, the specific type of material can be accurately identified. Once the material is identified, we can obtain its hardness and other relevant data by referencing existing databases.

3.2.2 Material Verification for Wood

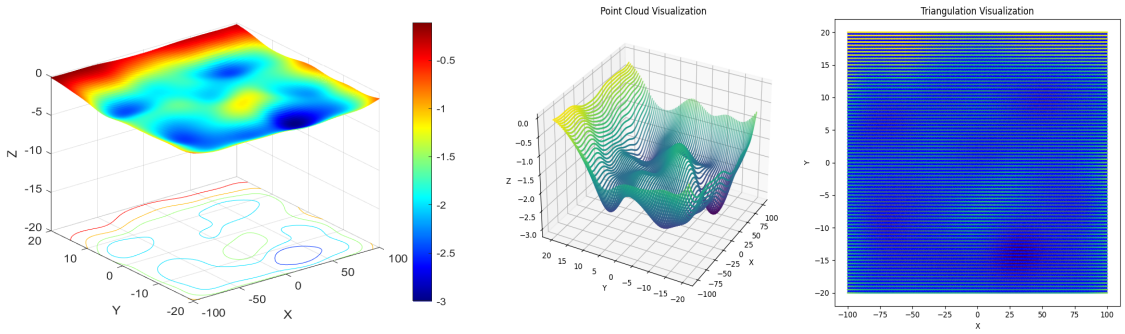
Since different types of wood have distinct grain patterns, colors, and textures, a trained observer can identify these textures and estimate the wood species through direct visual inspection. We can obtain relevant material characteristics from existing databases.

3.3 Wear Calculation

Wear volume (W) is calculated by processing raw point cloud data from a hand-held LiDAR scanner through 3D reconstruction into a surface model. The depression volume is determined using numerical integration with Delaunay triangulation and linear interpolation to map the irregular point cloud onto a regular 2D grid. The volume of wear is calculated using a specific formula:

$$W = \sum_{i} (z_i \cdot \Delta x \cdot \Delta y) \tag{1}$$

where z_i is the interpolated height value at grid point (x_i, y_i) (if $z_i < 0$); Δx and Δy are the grid spacings in the x and y directions; The summation is performed over all grid points where $z_i < 0$.



(a) Point Cloud Visualization (b) Volume Visualization

Figure 3: A Example of Wear Calculation

4 Strain Process and Stress Distribution Analysis

We develop Strain Process and Stress Distribution Analysis for two primary objectives. First, to establish a mathematical model and conduct simulation analysis of the stair structure by **Finite Element Analysis (FEA)**. Second, to investigate the stress, deformation, and dynamic behavior of the stairs under varying load and environmental conditions.

4.1 Mathematical Modeling of Different Stairs

FEA is used to address the analytical complexity of staircases with diverse geometries(as shown in Figure 4 and Figure 5). This approach models the staircase via discretization and meshing, translating mechanical interactions (e.g., friction, compressive stresses, and shear forces) into mathematical equations. Numerical methods are applied to solve the model and efficiently describe its mechanical behavior. By approximating the staircase with standardized small segments, its physical structure is converted into a mathematical representation, covering the following aspects:

(1) Discretization Approximation Structure: The staircase is discretized into finite elements (FEM) to approximate its continuous structure. Each step is treated as a **fundamental unit**, solved using differential equations. This enables unified analysis of various staircase types, with adjustable parameters like height, width, and depth. Uniform materials and connections are applied, with specific adjustments for cases like spiral stairs, where the angle θ optimizes force distribution.

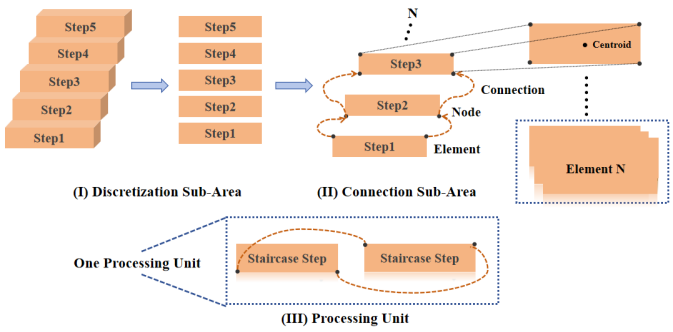


Figure 4: Standard Staircase (Basic): ***The continuous ascending space is discretized into equally spaced steps, where each step functions as a "discrete unit".***

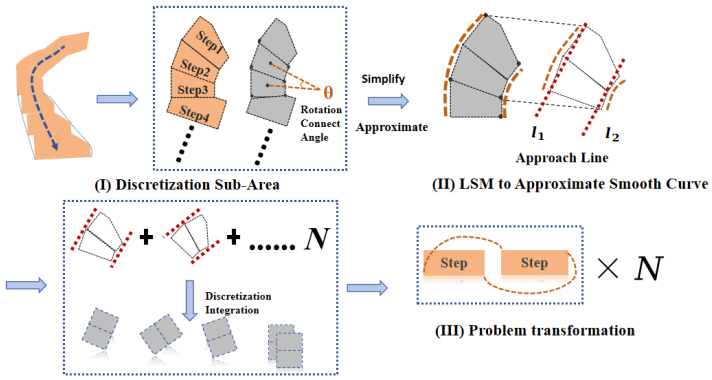


Figure 5: Spiral Staircase: ***The spiral path follows a curve rotating around a central axis. For analysis, it is discretized into multiple small step segments. Each segment is analyzed individually using FEA , and the sub-results are combined to approximate the structuralbehavior of the entire staircase.***

(2) Quantification of Initial Conditions: The initial physical state of the stair materials, including geometry, material properties, load levels, and the pressure-stress relationship, must be set. These conditions are crucial for analyzing the stairs' response to external factors like wear, vibration, and load changes. These attributes are all tested and listed on the Table 2.

(3) Force Balance Equation: To describe the equilibrium state of the staircase under load, the short duration of the applied load can be considered as instantaneous, allowing the problem to be treated as static. In this context, the deformation of the staircase can be approximated using the equations of elastic mechanics.

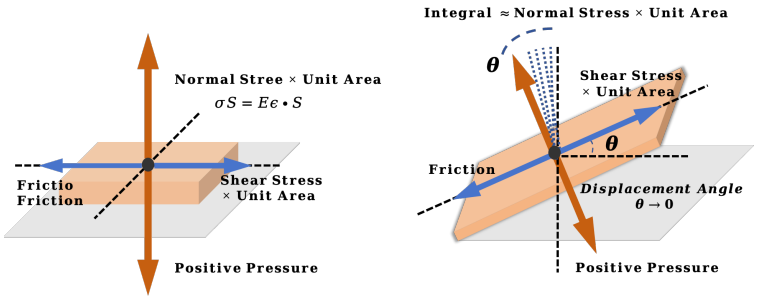


Figure 6: Schematic Diagram of Force Analysis on Stairs

4.2 Partial Differential Equation Analysis of Force

After modeling the staircase, **Partial Differential Equations (PDEs)** are used to analyze stress, deformation, and dynamic response. External loads cause stress and strain, with the load-strain relationship described by elasticity equations. In plastic material models, when stress exceeds the yield strength, the stress-strain relationship becomes nonlinear in the plastic deformation regime.

$$\nabla \cdot \sigma + f = 0 \tag{2}$$

where σ is the stress tensor, representing internal forces in the material; f is the body force (e.g., gravitational or internal); $\nabla \cdot \sigma$ describes how stress is distributed.

When stress exceeds the yield strength, the stress-strain relationship becomes nonlinear. In a ductile material model, the equation between stress and plastic strain is:

$$\sigma_{ij} = \sigma_{ij}^{el} + \sigma_{ij}^{pl} \tag{3}$$

where σ_{ij}^{el} represents the elastic stress component, which can be calculated using **Hooke's Law**. σ_{ij}^{pl} denotes the plastic stress component, typically **associated with plastic strain and yield criterion**.

In plastic deformation, the stress-strain relationship follows the von Mises yield criterion, which defines the material's yield condition. When stress exceeds the critical point, the material enters the plastic regime, causing plastic strain. The von Mises yield criterion is expressed as:

$$\sigma_v = \sqrt{\frac{1}{2}\left(\sigma_{ij} - \frac{1}{3}\text{tr}(\sigma)\delta_{ij}\right)^2} = \sigma_p 0 \tag{4}$$

where σ_v is the von Mises equivalent stress; $\text{tr}(\sigma)$ represents the trace of the stress tensor, indicating the mean stress;σ_p denotes the yield strength.

Strain and Displacement Fields reflects the relationship between the displacement at each position point and the strain at a specific point.

$$\epsilon = \nabla u \tag{5}$$

where ∇u is the gradient of the displacement field u, representing the spatial rate of change of displacement.

By solving the partial differential equation with boundary conditions and external loads, the stress and deformation distribution of the staircase under static loading can be obtained.

Second, for dynamic loads (e.g., wear, vibration, varying loads), the **Kinetic Equations (KE)** are used to account for external forces. This analysis considers mass, damping, material stiffness, and the effects of **material properties** (e.g., elastic modulus and Poisson Ratio) and geometric parameters (e.g., cross-sectional area) on **deformation**.

The Finite Difference Method (FDM) is used to solve KE through numerical discretization, approximating the solution over discrete time steps. The dynamic behavior is described by a second-order ordinary differential equation:

$$M \cdot \frac{u_{n+1} - 2u_n + u_{n-1}}{(\Delta t)^2} + C \cdot \frac{u_{n+1} - u_{n-1}}{2\Delta t} + K \cdot u_n = L(t_n) \tag{6}$$

where M is the mass matrix (mass distribution), C is the damping matrix (energy dissipation), K is the stiffness matrix (resistance to deformation), and $F(t_n)$ is the time-varying external load vector (e.g., seismic waves, human activity).

FDM is used to solve the kinetic equation by discretizing time derivatives like acceleration and velocity, approximating the solution over discrete time steps.

4.3 Discretization and Finite Element Method(FEM)

Based on the discretization of the previously described staircase physical model, linear elements were used for modeling. Each segment of the staircase is treated as a separate discrete sub-element, and all sub-elements are connected through nodes to form a mesh.

$$\int_{\Omega} \sigma(u) : \delta u \, d\Omega = \int_{\Omega} f \cdot \delta u \, d\Omega + \int_{\partial\Omega} t \cdot \delta u \, dT \tag{7}$$

where δu is the virtual displacement, t is the boundary load, Ω is the domain, $\partial\Omega$ is the boundary.

Next, we conducts simulation experiments on nine staircase steps made of the same material (with identical and uniform physical properties) under different loading conditions.

Table 3: Simulation Experiment Details

Component	Description
(a) Sample	Staircase steps (step 1 to 9) made of a material with uniform strength, initial stiffness, and integrity.
(b) Variable	The load increases linearly and steadily with the step sequence number ($\Delta L = m$).
(c) External Noise	Physical (weathering and accumulation), chemical (aging and decay) and biological (microbial attachment)
(d) Control Data	The stiffness matrix (K) reflects the staircase structure's ability to deform under applied loads, with a linear relationship $L = K \cdot u$ (where u is displacement and L is the external load).

For each staircase step, the load (L) is calculated, and Gaussian noise is added to simulate load disturbances. (K) is assembled, and the load is applied. We can generate an image of deformation displacement changes under varying external loads.

The results reveal deformation displacement deviations in steps 2, 6, 7, and 8, indicating varying wear levels under environmental disturbances. This enhances the model's adaptability to environmental changes and nonlinear behaviors, improving regression prediction robustness and effectively addressing nonlinear deformations and responses.

The conclusion is that for a specific material's stiffness K, the deformation displacement is linearly related to the applied load, expressed as $u = \sum_{0}^{n\times\omega} L(\theta) \cdot K^{-1} + m$. Through this equation, the required load for different materials (each with unique stiffness K, elastic modulus E, and Poisson Ratio ν) under a specific deformation displacement Δs can be determined.

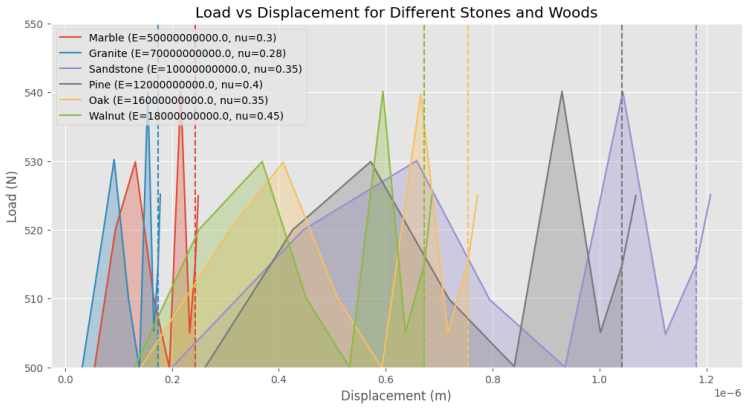


Figure 7: Load vs Displacement for Different Stones and Woods: ***For a material to undergo slight deformation, a minimum load threshold is required.***

5 Part 1: Usage Frequency and Age Estimation Problem

5.1 Usage Frequency Model

5.1.1 Derivation of External Load based on Wear Volume

The Equivalent Elastic Modulus (EE) is a single value used to characterize the overall elastic behavior of a complex material or structure simplified as homogeneous. The equation for calculating the Equivalent Elastic Modulus (EE) is as follows:

$$E^* = \frac{E}{1 - nu^2} \tag{8}$$

First, we model the contact area and time duration to analyze the interaction dynamics and their impact on the system's behavior.

Humans primarily have several types of foot shapes, including Square Foot, Celtic Foot, High Arches, and Low Arches. To facilitate the modeling process, we simplify these various foot shapes into an elliptical form , as shown below. The major and minor axes of the ellipse are based on the average shoe size of the population.

Research shows that humans typically use the forefoot to land and exert force when moving on stairs. Thus, the contact area is defined as the overlapping region where the forefoot interacts with the stair steps through friction and compression.

Table 4: Statistical survey on foot size in some countries

Country	American		European		China		Japan		Korean	
Shoe Size	M:11 US	F:8 US	M:43 EU	M:39 EU	M:41 CN	M:38 CN	M:27 JP	M:23 JP	M:274 KR	M:242 KR

The approximate major and minor axes of the ellipse are simplified and estimated to obtain the equivalent contact radius R.

$$R = \left(\frac{7a}{6} \times \frac{11b}{10} \right)^{\frac{1}{2}} \tag{9}$$

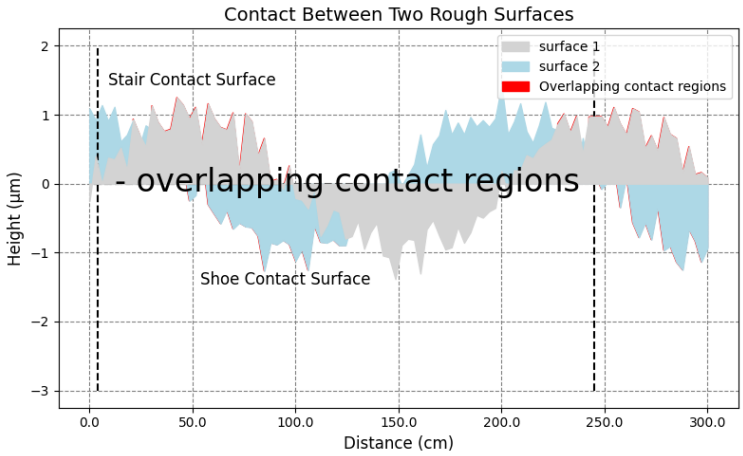


Figure 8: Contact Between Two Rough Surfaces: ***Through experimental simulation, we determined that the range of the primary force is concentrated within the interval of 0 to 247.8329 cm, thereby defining the area of force application.***

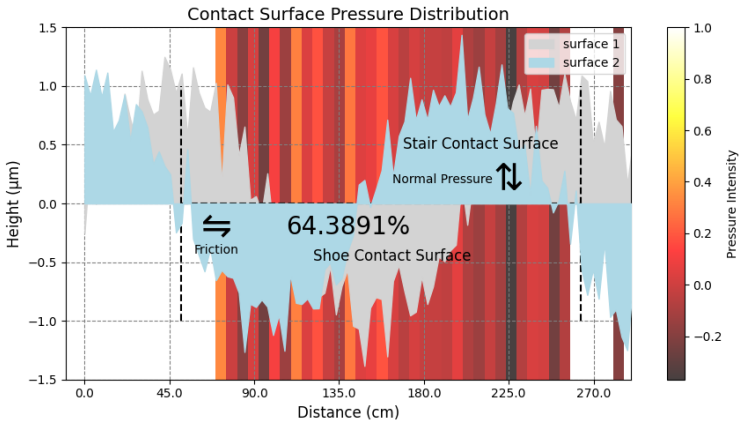


Figure 9: Contact Surface Pressure Distribution: ***Primary contact forces during foot-stair interaction cover 64.3891% of the sole area. Uniform thermal distribution indicates stable force application, with*** $S_{Force} \approx S_{Contact}$.

The contact time between the human foot and the stair step is very brief. Assuming constant speed and uniform step length while ascending, the contact area with each step remains equal, resulting in a fixed relative frictional distance during each interaction. Therefore, the contact time with each step is constant. We use Naismith's rule to estimate climbing time and calculate the average speed.

We estimate the time per step climb $t_{\text{unit_climb}}$ using Naismith's rule.

$$t_{\text{unit_climb}} = \left(\frac{S_{\parallel}\,(\text{m})}{5000}\right)^{1.3} + \left(\frac{S_{\perp}\,(\text{m})}{600}\right)^{1.5} \tag{10}$$

Using the empirical rule, we determine the time for each foot contact with a stair step and then construct the Contact Mechanics and Stress Distribution Model.

Contact Mechanics studies the deformation of solids in contact, focusing on stress and strain when walking on stairs. The elastic and deformable rubber shoe sole ensures elastic deformation in the contact area. Using **Hertzian theory**, we describe **the relationship between contact forces** (normal pressure, adhesion, tangential friction) and **stair deformation**, which depends on factors like Contact Force and the deformation effects in the contact area.

For the contact force between two circular objects:

$$F = \frac{4}{3} \cdot \frac{E^* \cdot \delta^{3/2}}{R} \tag{11}$$

where E^* is the effective elastic modulus of the contact materials, δ is the deformation, and R is the radius of the contacting objects.

The wear of stair treads caused by shoe soles is a combination of adhesive wear and fatigue wear, characterized by multi-asperity contact. When rubber comes into contact with the surface of the material, friction can lead to localized adhesion phenomena. These adhered tiny stone particles are peeled off from the stair surface. Repeated friction and stress may cause microscopic cracks to form on the stone surface, but these cracks are typically localized and do not lead to brittle fracture.

Hardness and Wear Rate: There exists a quantitative relationship between material hardness, elastic modulus (E), and Poisson Ratio (ν).

$$H = \frac{\sqrt{E} \times \log E^*}{1 - nu^2} \tag{12}$$

Wear Rate: $k \times \frac{F}{H} + b$, where k is the wear coefficient, F is the contact force, H is the hardness, and b represents external environmental influences.

The wear coefficient is influenced by factors like friction conditions, surface roughness, and load, meaning it changes during human motion on stairs. A literature review helps determine the range and variation process of the wear coefficient for materials commonly used in stair construction.

Table 5: Selected wear coefficient ranges for three common materials under normal conditions.

Material	Minimum Wear Coefficient	Maximum Wear Coefficient
Marble	0.119284	0.548221
Sandstone	0.310473	0.728493
Pine	0.402948	0.679301

From Figure.10, by fitting the variation curves of the wear coefficients for different materials, it was determined through multiple fittings that logarithmic fitting provides the best approximation for the decay curves. Therefore, logarithmic fitting was selected to simulate the decay of the wear coefficients.

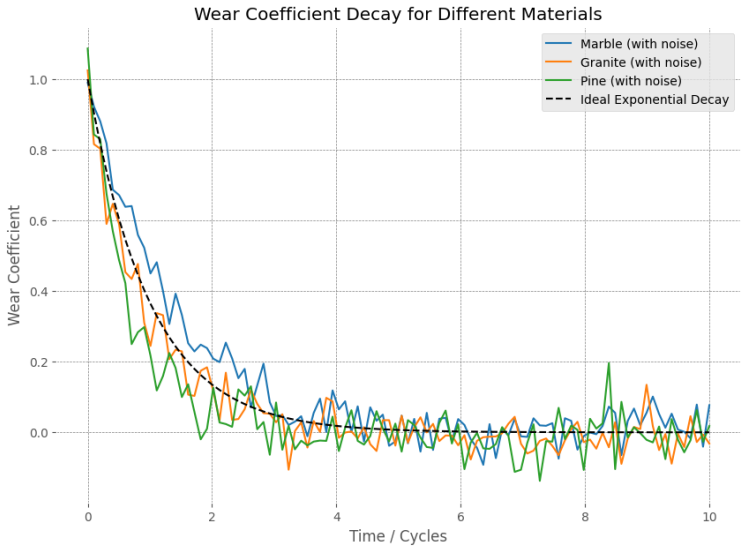


Figure 10: Wear Coefficient Variations of Stair Materials Under Three Standard Conditions

$$k(t) = k_{\max} - \frac{(k_{\max} - k_{\min})}{\log(N(F,t))} \times \log n(F,t) \tag{13}$$

Using the **Archard Equation**, environmental noise (b) is added to account for the influence of external conditions on stair wear, combined with the wear coefficient $k(t)$ that varies with environmental effects. Then we can get wear rate:

$$Q = k(t) \times \frac{F}{H} + b \tag{14}$$

where $k(t)$ is the wear coefficient, F is the contact force, H is the hardness, $b(t)$ represents the influence of external environmental conditions.

To ensure that the noise term $b(t)$ reflects the influence of external factors (physical, chemical, and biological factors), the noise is represented as a combination of multiple factors and linked to environmental variables such as temperature, humidity, weathering, etc.

$$b(t) = \int \mu(t) + \sigma(t) \cdot (\epsilon_t + \alpha_1 T(t) + \alpha_2 H(t) + \alpha_3 B(t)) \tag{15}$$

where $\mu(t)$ represents the baseline noise (independent of time), $\sigma(t)$ denotes the noise intensity (varying with time), $\epsilon(t)$ is white noise representing random environmental disturbances, $\alpha_1, \alpha_2, \alpha_3$ are the influence coefficients of different environmental factors, and $T(t)$, $H(t)$, $B(t)$ represent the environmental temperature, humidity, and microbial activity at a specific time t, respectively.

We can obtain the equation of wear rate Q through the equation 13, 14 and 15.

Next, differential equation describes how wear W(t) changes over time t.

$$W(t) = \int_0^t \left[\left(k_{\max} - (k_{\max} - k_{\min}) \cdot \frac{\log(F,t)}{\log(N(F,t))} \right) \cdot \left(\frac{F}{H} \right) + b \right] dt \tag{16}$$

Through multiple experimental simulations, considering the influence of environmental noise, the time required for the wear of staircase material volume according to Hertzian theory was simulated.

In this simulation of Figure.11, a specific construction material (with constant stiffness K, hardness H, elastic modulus E, and Poisson Ratio ν) was used to study the relationship between total wear volume and wear time. Since the same material and contact force were applied, the wear volume of the staircase within a certain range corresponds to a specific wear time interval t. The observed oscillations reflect the influence of external environmental factors. It also helps assess the extent of environmental impact.

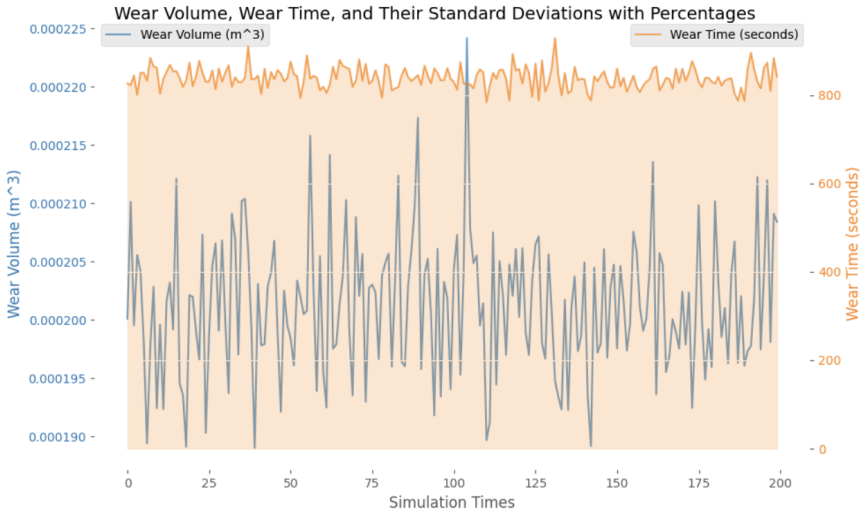
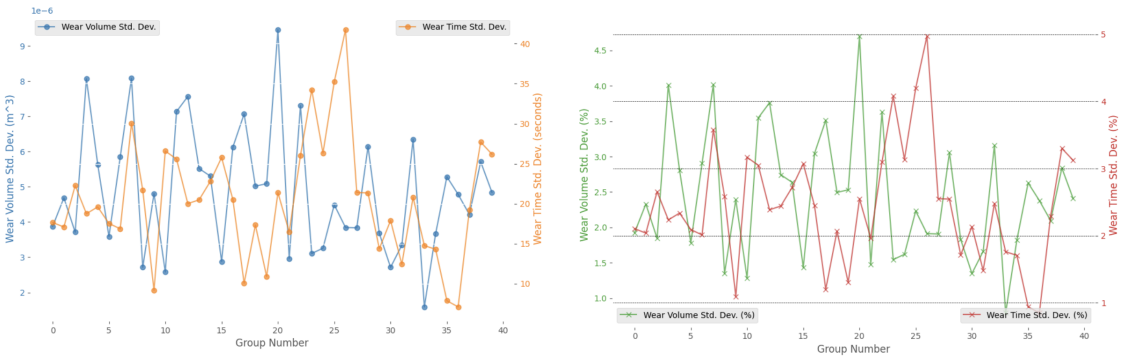


Figure 11: Staircase wear volume vs. time in simulations (with environmental factors)

In Figure.12a and Figure.12b, the simulation experiment was evaluated, and the results indicate that the model's estimated error for the required wear time t, derived from the input of staircase material wear volume, is less than 4.5%. Most errors remain stable within the range of 1.3742% to 3.8921%, demonstrating the model's good stability.



(a) Standard Deviation of Staircase Wear Volume vs. Time in Simulations

(b) Percentage Error of Staircase Wear Volume vs. Time in Simulations

Figure 12: Simulation Results for Staircase Wear Analysis (with Environmental Factors)

5.1.2 Usage Frequency Based on Wear Volume and Climbing Time

We use the relationship between wear volume (W) and the time required for material wear (t_{wear}) along with Naismith's rule to obtain climbing time. This allows us to estimate the average

duration of interaction between each footstep and the ground ($t_{\text{unit_climb}}$). By assuming a constant average speed, we can estimate how many times the staircase has been used since its construction. This approach also extends from individual steps to the entire staircase.

First, the total wear volume (W_{current}) is measured using an instrument.

Based on the a forementioned theory of elastic contact mechanics, the analysis of material properties, and the formulas in the above content, we can estimate the time required to achieve this wear amount:

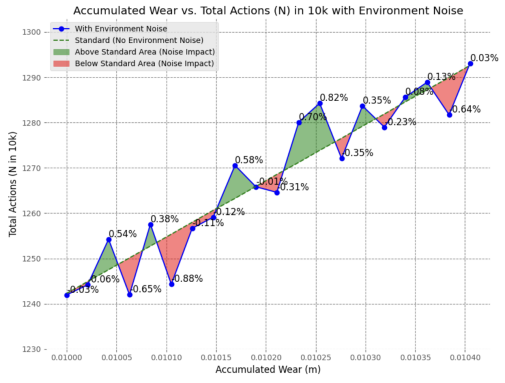
$$t_{\text{cost}} = \frac{\int_0^t \left[k(t) \times \left(\frac{F}{H}\right) + b(t)\right] \, dt}{k_{\max} - \left[(k_{\max} - k_{\min}) \times \frac{\log(F,t)}{\log(N(F,t))} \times \left(\frac{F}{H}\right)\right] + \sum_0^t b(t)} \tag{17}$$

Finally, based on the assumption that people ascend the stairs at a constant speed, with the same average load, and with the same step pace," we can determine the number of actions N required for the stair steps to reach the specified wear amount W_{current}.

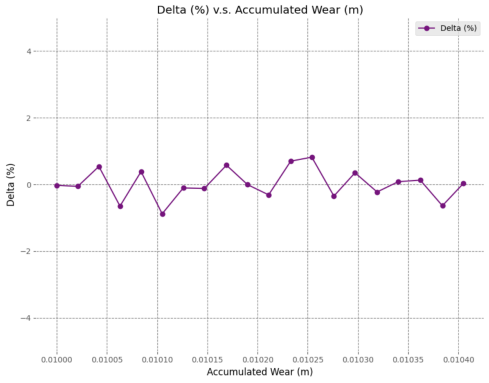
$$N = \frac{t_{\text{cost}}}{t_{\text{unit_climb}}} = \frac{\int_0^t \left[k(t) \times \left(\frac{F}{H}\right) + b(t)\right] \, dt \times \left[\left(\frac{S_{\parallel}(m)}{5000}\right)^{1.3} + \left(\frac{S_{\perp}(m)}{600}\right)^{1.5}\right]}{k_{\max} - \left[(k_{\max} - k_{\min}) \times \frac{\log((F,t))}{\log(N(F,t))} \times \left(\frac{F}{H}\right)\right] + \sum_0^t b(t)} \tag{18}$$

Through a series of simulated experiments, we investigated the impact of varying wear depths on the frequency of stair usage under conditions of gradually increasing wear volume, and obtained corresponding experimental data.

Total Wear (cm)	1.0000	1.0063	1.0105	1.0147
Total Loading Actions (10K)	1242.2963	1250.1149	1255.3532	1260.6145
Times of Pedestrians Use (10K)	1241.7428	1249.9014	1255.1921	1260.3102



(a) Total Wear vs. Total Actions in 10k with Environment Noise



(b) Delta Percentage Change vs. Accumulated Wear Distance (m)

Figure 13: Interaction of Usage Frequency and Environmental Fluctuations in Stair Wear

The analysis reveals the frequency of use corresponding to a specific wear amount. While a linear relationship was observed in simulations, real-world scenarios show fluctuations due to external environmental factors. These fluctuations are represented by color-filled regions, with

larger areas indicating greater environmental influence. Despite these variations, the overall wear process is primarily driven by human activity.

5.1.3 Usage Frequency Calculation of Repaired Staircases

Meanwhile, considering that the stairs have undergone repairs, the number of repair and renovation instances can be confirmed based on the 3D reconstructed model and the layered structure obtained from the Ultrasonic Thickness Gauge.

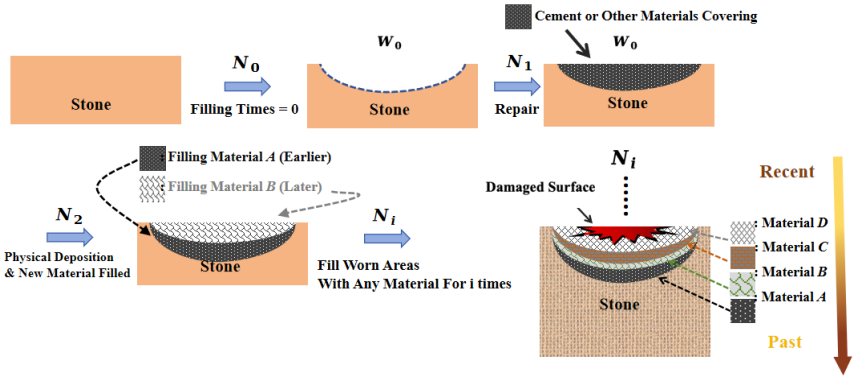


Figure 14: Demonstration of the Historical Wear Progression of a Repaired Staircase: ***A, B, C and D are made of the same material, and it is assumed that there is no friction between the materials, with complete filling.***

From the equation 18, we can get the equation of W_{current}.

During the first repair, the wear volume of the stone is W_0 (at this stage, the wear is caused by shoe soles). After the first repair and N_1 usage cycles, the total wear volume of the stone is $W_{1\text{total}}$. The wear volume of the repair material after N_1 cycles is ΔW_1. Therefore, the wear volume of the stone during the N_1 cycles is calculated as (the wear is caused by repair materials):

$$W_1 = W_{1\text{total}} - \Delta W_1 - W_0 \tag{19}$$

By extension, the wear volume for the i-th cycle is:

$$W_i = W_{i\text{total}} - \Delta W_i - W_{i-1}, (W_{i\text{total}} = W_{\text{current}}) \tag{20}$$

Using LiDAR, we can determine the number of repair layers, indicating how many times repairs have been conducted. Through 3D reconstruction and processing, we can quantify the wear volume for each layer, i.e., the value of $\Delta W_i - \Delta W_{i-1}$. Consequently, the total number of usage cycles N_{total} can be derived as:

$$N_{\text{total}} = \sum_{i=0}^{N} N_i \tag{21}$$

5.2 Age Estimation Model

5.2.1 Age Estimation Model Based on Wear Volumn

We utilizie the **Poisson distribution** to model the frequency of stair usage and employ **Monte**

Carlo simulations to account for the inherent variability and randomness in human flow patterns. The dynamic environmental conditions are factored into the simulation to reflect their influence on the wear rate of the staircase material over time. By dividing the total volume of the staircase by the mean wear rate obtained from the simulations, we derive an estimation of the staircase's age.

The equation is as following:

$$A = \frac{\sqrt{k \times \frac{F}{H} + C \cdot \sigma_{\text{contact}}^{n}} + b(t) \cdot e^{-1}}{\left(k_{\max} - \left[(k_{\max} - k_{\min}) \times \frac{\log((F,t))}{\log(N(F,t))}\right] \times \left(\frac{F}{H}\right)\right) + b(t)} \tag{22}$$

The model estimates step wear volume over time, incorporating environmental factors and pedestrian flow. It calculates the time for a step to erode to a specific wear volume, representing its age. Granite shows the longest age at the same wear volume, with error rates of 3.89%–5.29%. Steel achieves the best prediction accuracy (2.75% error). For a wooden stair in a dry, cold environment with stable pedestrian flow, the estimated age confidence interval is [238.13, 251.10] years, demonstrating the model's reliability.

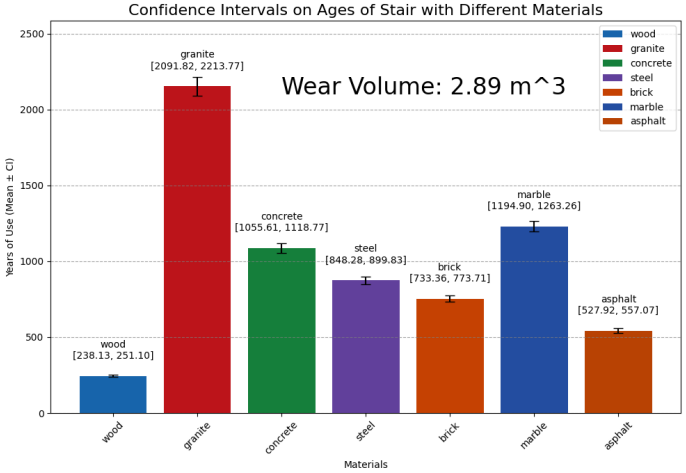


Figure 15: Confidence Intervals on Ages of Stair with Different Materials

5.2.2 Age Estimation Model Based on Weathering Rind

Weathering refers to chemical and physical changes over time, with rind development varying by region and conditions, typically correlating with time. McSaveney (1992) proposed weathering rind thickness logistics model, but no suitable model was found despite fitting attempts. Regression analyses showed high explained variances, but models either failed to align thickness with time or predicted unrealistic infinite growth.

Consequently, we adopt **a logistic model** for the data, which consists of the weathering rind thickness and years for the same material under the same environmental conditions, to fit the parameters of the above equation.

$$f(t) = \frac{a}{1 + e^{b(t-c)}} + d \tag{23}$$

where a represents the range of the weathering rind property across the chronofunction, from $t = -\infty$ to $t = +\infty$, b is the slope coefficient, c denotes the time (in years) at which the rate of change is maximal, and d is the asymptotic value as t approaches $+\infty$.

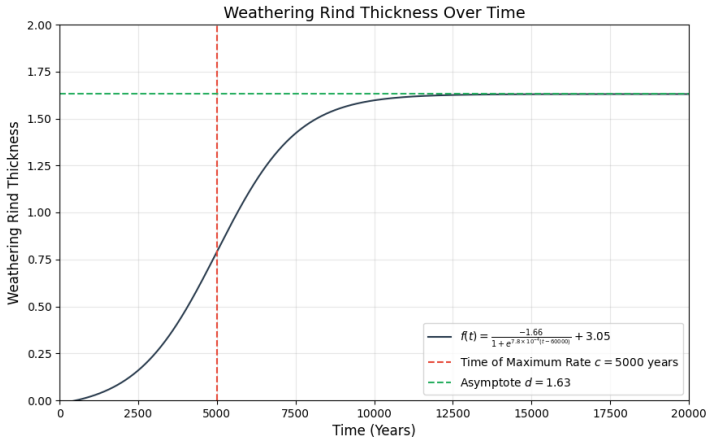


Figure 16: Weathering Rind Thickness Over Time (An Example)

If we know the thickness of weathering rind, we can get the age of the stair. Since we have already identified the material of the stairs, we can consult relevant literature to fit a curve and determine the parameters in the formula. Finally, we can get an accurate age.

5.2.3 Combined Age Estimation via Weighted Averaging

To integrate the two models, a weighted averaging approach is employed. Specifically, let A_{wear} denote the age estimate derived from the Poisson Distribution and Monte Carlo simulation, and $A_{\text{weathering}}$ represent the age estimate obtained from **the weathering rind model**. The combined age estimate, A_{combined}, is formulated as follows:

$$A_{\text{combined}} = w_1 \cdot A_{\text{wear}} + w_2 \cdot A_{\text{weathering}} \tag{24}$$

where w_1 and w_2 are weighting coefficients that quantify the relative reliability of each model.

5.3 Usage Statistics and Temporal Distribution Analysis

In the preceding section of our study, the total number of times the staircase is used and the construction time have been calculated. Using this data, we can estimate the number of times the staircase is utilized per day, which in turn allows us to determine the daily number of users.

$$N_{\text{daily}} = \frac{N}{A_{\text{combined}}} \tag{25}$$

Miner's Rule states that fatigue damage accumulates linearly with each load cycle. If the model's construction time exceeds known data, it suggests less frequent usage over a longer period, resulting in gradual wear and slower fatigue accumulation. Conversely, a shorter calculated time implies intense, concentrated use, leading to higher local stresses and accelerated fatigue, which increases the risk of structural failure.

6 Part 2: Stairs Usage Pattern and Pedestrian Flow Problem

6.1 Direction Preferences

Through the point cloud formed by LiDAR scanning from the site of the archaeological stair-

case, we can get the distribution of the wear centers and observe that the step plane has obvious zoning.

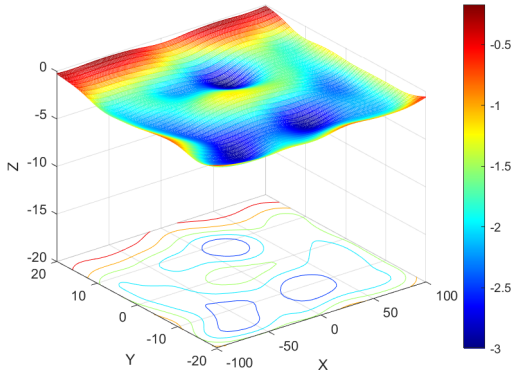


Figure 17: Point cloud example

It is important to note that: All the point cloud diagrams mentioned in this paper represent a single step of the target staircase: the line $y = h, z = 0$ corresponds to the inner edge of the step, while the line $y = -h, z = 0$ corresponds to the outer edge. With the observation direction set as going upstairs, the line $x = -\frac{l}{2}, z = 0$ represents the left side, and the line $x = \frac{l}{2}, z = 0$ represents the right side.

6.1.1 From a horizontal perspective

When going upstairs and downstairs, the sole of the front foot touches the steps, with metatarsal heads(a point on the palm of the forefoot) as the landing point. The wear on the inner edge(zone B) of the landing area is caused by going upstairs, and the wear on the outer edge(zone A) is caused by going downstairs.

The distribution of the wear centers are mainly divided into zone A and zone B by observing the point cloud.

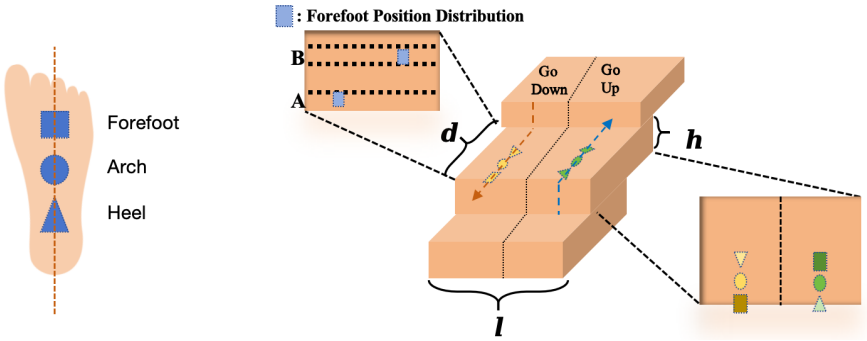


Figure 18: Force points on stairs

Therefore, zones A and B can show the specific direction of the people who used the stairs in history(assumption: The impact of landing on the heels on the steps is minimal and can be ignored).

The foot begins from a stationary position and then free fall with the speed of the foot increases from 0 to v_t and when it touches the surface of the next step, its speed rapidly decreases to 0 m/s. The speed diagram is as follows:

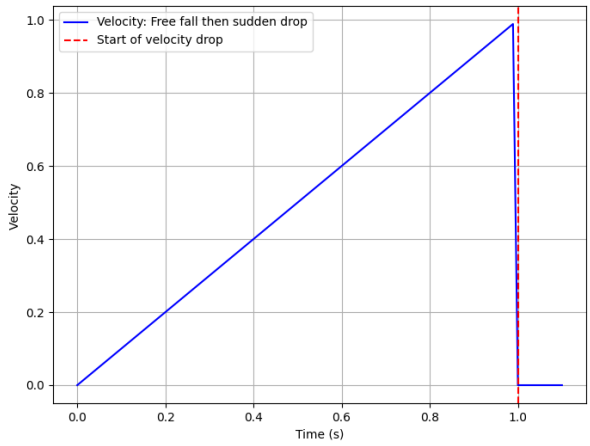


Figure 19: The velocity variation of going downstairs

The dynamic and mechanical analysis of going downstairs are as follows:

$$v_t^2 = 2gh \quad (26) \qquad p = mv_t \quad (27) \qquad F_{\text{Impact}} = \frac{\Delta \mathbf{p}}{\Delta t} \quad (28)$$

where the time taken for the speed to drop rapidly to 0 m/s is Δt = 0.1 s.

$$F_1 = F_{Impact} = 10m\sqrt{2gh} \tag{29}$$

The action of lifting the leg and stepping onto the step is accomplished by the control of muscles, so the speed of the foot on next step is 0 m/s.

The mechanical analysis of going upstairs is as follow:

$$F_2 = mg \tag{30}$$

In order to verify statistically the difference of wear in different regions, the confidence interval method was adopted. The null hypothesis H_0 is defined as:

$$H_0 : \frac{W_A}{\frac{F_1}{F_2}} - W_B = 0 \tag{31}$$

where the triangulation and interpolation methods provide the wear volumes W_A and W_B for zones A and B.

We generally believe that significance level = 0.05. $\frac{W_{\text{A}}}{k} \gg W_{\text{B}}$ indicates that the primary walking pattern for the target staircase is going upstairs. Conversely, the primary walking pattern is going downstairs.

6.1.2 From a vertical perspective

The minimum width required for walking is slightly larger than the shoulder width of an individual. Based on anthropometric data, we adopt the larger shoulder width of 0.4 m as the standard required width for an individual's walking space.

When the width of the staircase is less than twice the larger shoulder width of 0.8 m, we can think that the staircase is relatively narrow, then it will be found that the two most worn points are in the same straight line (one in the inner edge and one in the outer edge), people walk on the target staircase in a single row, and then the width of the staircase l is divided into three parts to determine which area of the left, right, and middle of the two points are connected, and it is believed that people tend to go which side.

For non-one-way streets, we are generally divided into left and right. So we have to find out the point where the deepest wear(d_w) is, determine whether it is on the left or right side of the middle line of the stairs, which way people tend to go.

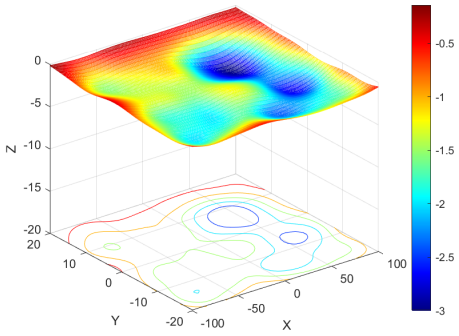
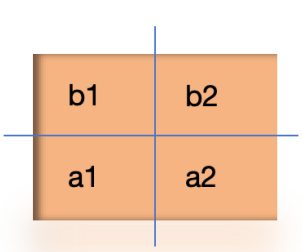


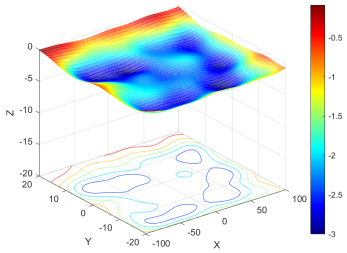
Figure 20: Point cloud map which show that prefer walking on the right side

6.1.3 Conclusion of the direction

If stairs are used frequently, people generally tend to follow the crowd and adopt the same walking preferences, thereby establishing a distinctive walking pattern for the target staircase.



(a) Abstraction of Partitioning



(b) Point cloud example of the Left-up, Right-down pattern

Figure 21: Determine walking pattern by point cloud

Table 6: The features of point cloud with different usage tendencies

Walking pattern	Description of point cloud (relationship of wear in four regions)
Left-down, Right-up	biggest: a1; smallest: a2
Left-up, Right-down	biggest: a2; smallest: a1
Right-up, Right-down	$a2 > b2 \gg a1, b1$
Left-down, Left-down	$a1 > b1 \gg a2, b2$

6.2 Simultaneous Perferences

We can know the walking pattern and transverse wear distribution from the point cloud, and measure the wear width(l_w). l_w is the total width of all the areas of the deepest wear height.

$$n_{sbs} = \frac{l_w}{l_{\text{Shoulder}}} \tag{32}$$

where n_{sbs} represents the number of people walking side by side, with the condition that they are walking in the same direction. The standard required width for an individual's walking space, $l_{\text{Shoulder}} = 0.4$ meters.

7 Sensitivity Analysis

We have constructed two core models: the **Stair and Mechanical Analysis Model (SEMA)** and the **Stair and Environmental Factor Analysis Model (SEFAM)**, which deal with the tasks of stair and mechanical analysis and stair and environmental factor interaction, respectively.

In SEMA, we focus on mechanical factors like load intensity and frequency, and material properties (stiffness, hardness, elasticity). To assess "What most influences staircase wear?", four parameters are analyzed: **1) material properties**, **2) load intensity**, **3) load frequency**, and **4) walking pattern**. The oscillation range is 5–7%, and Monte Carlo simulation is used for sensitivity analysis. Results show that with fixed material properties and load intensity, increasing load frequency from 0.5 to 1.5 causes a significant increase in wear (from $\Delta W = 2.9301 \times 10^{-4} cm^3$ to $\Delta W = 3.6491 \times 10^{-4} cm^3$), with a 24.54% change, indicating that frequency significantly affects wear. Thus, in SEMA, usage frequency should be given more weight for model stability.

For SEFAM, environmental factors like temperature, humidity, weathering, and human restoration are considered. For the question "How does the environment affect wear?", the four parameters analyzed are: **1) temperature**, **2) humidity**, **3) weathering**, and **4) restoration**. The parameter oscillation range is also 5–7%. Monte Carlo simulation shows that stable wood properties and consistent mechanical conditions result in significant wear increases: When **factor 1)** the ambient temperature increases linearly from to, the wear of the wood stairs increases significantly $\Delta W = 2.2389 \times 10^{-4} cm^3$. When **factor 2)** the ambient humidity increases linearly from RH=30% to 50%, the wear of the wood stairs also increases significantly $\Delta W = 3.3215 \times 10^{-4} cm^3$. Temperature and humidity have the most influence on wear, while weathering and restoration have less impact. Therefore, in SEFAM, temperature and humidity should be prioritized for model stability.

8 Strengths and Weaknesses

8.1 Strengths

1. Finite Element Analysis: Finite element analysis methods provides accurate simulation results for research and can help optimize designs and predict performance.

2. Visualization techniques are used: We visually display the model's structure, results, and change trend, which enhances the comprehensibility and operability of the research.

3. Environmental factors are considered, which enhances the robustness of the model: We fully consider external environmental factors, such as temperature, humidity, pressure and other variables, when constructing the model, which provides stronger adaptability and robustness for the application of the model.

4. Reasonable cost control and budget allocation were arranged: Our program have rationally controlled the budget to ensure the efficient use of funds and optimal allocation of resources. This is crucial for cost management in real-world applications and helps to promote and apply the model.

8.2 Weaknesses

1. Application limitations: Although the model has considered some common environmental factors, there are still limitations. Other materials may have different specific properties that are not captured in the current model.

2. Insufficient generalization of the model: The current model may only be applied to specific types of materials or a limited number of engineering problems, and its generalizability and adaptability need to be improved.

3. Limitations of assumption conditions: The model's accuracy and applicability depend on the assumptions about environmental factors and material properties, which may not fully reflect reality.

9 Conclusion

In this study, we developed **3 different models** to deal with analysis corresponding to different aspects: **1) SEMA Model** is to simulate the relationship between **wear and physical mechanics**, and handle the process of stress and strain. **2) SEFAM Model** is to simulate the relationship between **wear and environmental factors**, and handle the process of various environmental influences on stairs. **3) SUP-PFM Model** is to focus on **flow analysis**, combining wear and tear with crowd characteristics (crowd distribution, crowd density). Thus our model successfully integrates key physical factors such as step geometry, material properties and external forces. However, the model's applicability is limited by certain assumptions, such as the uniformity of material properties and walking mode of people. These assumptions may not hold true in all real-world scenarios, which could affect the accuracy of the model in some cases.

In future work, it would be beneficial to refine the models by incorporating more complex material properties and environmental factors to enhance its accuracy and generalizability. Additionally, expanding the model to account for other architectural features could offer further opportunities for optimization. We hope our models can provide valuable insights into the further study of the wear of stairs and offers a path for further application in both academic and practical settings.

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Report on Use of AI

We used ChatGPT-4o mini to polish the language in our study, but all of the content were reviewed and revised by our team.

We also used ChatGPT-4o to help us understand the content in Problem A and Finite Element Analysis models. Both were manually evaluated.